Supply-Aware Sequential Matching: Round 2 Technical Report

Kashmira Matale
Team *sol*
24f3002508@ds.study.iitm.ac.in

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Challenge: Vouchsafe × IITM, Sequential Matching Problem (kit 1.0.0). Synthetic simulator only; nothing here is evidence about real people. Primary outcome: msmi per 100 arrived members, reported with calibration and who is left waiting.

1 Summary

In one sentence: the Round 2 policy matches greedy on the score but leaves clearly fewer people waiting for a first introduction; it does not prove a higher MSMI, and learned propensities do not reliably help.

Round 2 asked three questions. (1) The stdlib build reproduces the Round 1 prototype's decisions and, within noise, its results. (2) Shrunk member propensities, with priors recalibrated on selection seeds, do not show a reliable gain over a constant prior on fresh seeds; their ranking signal is weak (AUC 0.55, Brier equal to the base rate). (3) Under changed missingness and response behaviour, targeted constraint asks remain essential, soft-field asks never pay, and learning is at most weakly useful; the stress runs have only 4 episodes per condition, so this part is suggestive. The one change that held on both seed sets is a *waiting priority*: it reduces members left waiting and waiting time without a detectable change in msmi, mutual acceptances or dates against the supplied greedy baseline. It does trade away the mutual-acceptance gain that the Round 1 build showed pooled.

2 Process: how the work was done

1. **Starting point (Round 1).** Round 1 found that the supplied greedy baseline already introduces about 78% of the truly feasible pairs, so feasible supply, not allocation, caps the score. The Round 1 prototype (maximum-weight matching, learned propensities, targeted asks) looked better than greedy, but none of its ablations was separable from noise.
2. **Build and test the Round 2 policy.** I rebuilt the policy as one configurable, standard-library module (profiles `r1` and `r2`) and wrote an in-process paired evaluator that runs every policy on identical simulated worlds. Unit tests (9) check validity, determinism, exact matching against brute force, the fallback path and the waiting term.
3. **Selection stage (stage A).** 12 policy variants on 54 episodes: allocation (matching vs greedy), prior (constant vs learned), waiting weight $\lambda \in \{0, 1, 2, 4\}$, ask rule (targeted, baseline, none, soft), and an oracle using true probabilities as an upper bound. The waiting priority was the clearest change.
4. **Calibration.** Beta priors were fitted on stage-A feedback only, so the fresh seeds stay untouched.
5. **Fresh-seed validation (stage C).** The main candidates were rerun on 36 new episodes, together with the supplied no-clarification and random baselines and the Round 1 prototype. Results from both stages were then pooled (90 episodes).
6. **Changing conditions.** Five modified worlds: heavy missingness, low response, response dropping mid-episode, bimodal response, and strong member differences in acceptance (4 episodes each).
7. **Packaging and checks.** Official evaluator on seed 101 (all six scenarios valid, score 0.50), then the Docker container build and run (same result, about 0.8 s per call against a 10 s limit).

Changes of direction. I dropped soft-field asks (cost, no gain) and dropped learned propensities as the default (weak ranking signal, no confirmed outcome gain). I kept exact matching only because it is cheap, since greedy allocation does nearly as well once the waiting priority is in. The waiting priority was my only addition, and the one that held on fresh seeds.

Round 1 hypothesis	Status after Round 2	Evidence
H1 allocation (matching beats greedy)	Not needed once waiting is used	With the waiting priority, greedy allocation leaves -1.14 members waiting versus -1.24 for exact matching; no outcome difference.
H2 learned propensities	Weak, unconfirmed	Prediction AUC about 0.55, Brier equal to the base rate; pooled MSMI $+0.13$ $[+0.02, +0.26]$, not confirmed on fresh seeds alone.
H3 targeted asks save cost	Small saving; real value is shorter waits	About -3 ask units (about 2%), but mean wait -1.00 versus -0.23 with baseline asks.
H4 soft-field asks	Not supported	About $+51$ ask units with no gain on any outcome.

Table 1: Round 1 hypotheses revisited.

3 What changed

Round 2 default (`r2`). Every member has the same prior (no learned propensities); the daily batch is an exact maximum-weight matching (components ≤ 18 members, greedy beyond); an edge weight is multiplied by $1 + \lambda(s_a + s_b)$ with $\lambda = 2$, where $s_i = \min(\text{days since arrival}, 30)/30$ for members never yet introduced and 0 otherwise; matching relationship_goal (when both are observed) multiplies by 1.5; hard-constraint asks are targeted; no soft asks. **Round 1 build (`r1`):** learned Beta-shrunk propensities, no waiting term; kept as a profile. Hard eligibility always comes from the supplied `kit.eligibility`; outputs are re-validated and fall back to the supplied baseline on any error. The waiting term is my own definition; λ was not tuned beyond a sweep.

4 Protocol

All comparisons are paired on identical worlds, in-process, with the unmodified simulator (an exact memo of `eligibility` only speeds it up). *Stage A* (selection): seeds 3001–3009 $\times$ 6 scenarios = 54 episodes. *Stage C* (fresh, untouched during selection): seeds 6001–6006 $\times$ 6 = 36. *Stress*: seeds 7001–7004 in five modified worlds (70% of answers removed and 15% declined; response rates $\times 0.6$; response rate halved at day 30; bimodal response rate; strong member acceptance bias), 20 episodes. Intervals are 95% percentile bootstraps over paired episode differences (2000 resamples). "Left waiting" counts members with a truly feasible partner who never received an introduction; it uses simulator truth after the fact and is a diagnostic only. Calibrated priors were fitted by beta-binomial maximum likelihood on stage-A feedback only: responds $(0.78, 12)$, Yes $(0.45, 8)$, second-meeting Yes $(0.66, 50$, capped) versus hand-set $(0.8, 4), (0.4, 4), (0.53, 3)$.

5 Results

Policy (vs greedy)	Left waiting	Mean wait (d)	msmi/episode	Mutual acc.	Dates
Round 1 build	-0.30 $[-0.60, 0.00]$	-0.80 $[-0.99, -0.60]$	$+0.11$ $[-0.16, +0.38]$	$+0.66$ $[+0.02, +1.31]$	$+0.61$ $[-0.10, +1.40]$
Constant prior, no waiting	-0.23 $[-0.58, +0.12]$	-0.79 $[-1.00, -0.59]$	$+0.11$ $[-0.12, +0.34]$	0.00 $[-0.74, +0.76]$	-0.08 $[-0.81, +0.77]$
Round 2 (waiting)	-1.24 $[-1.52, -0.96]$	-1.00 $[-1.18, -0.81]$	$+0.08$ $[-0.19, +0.37]$	-0.07 $[-0.77, +0.63]$	-0.06 $[-0.83, +0.76]$
Waiting + learned	-1.27 $[-1.54, -0.99]$	-0.95 $[-1.13, -0.77]$	$+0.21$ $[-0.06, +0.47]$	$+0.09$ $[-0.53, +0.71]$	$+0.22$ $[-0.51, +0.96]$
Waiting, greedy alloc.	-1.14 $[-1.42, -0.88]$	-0.96 $[-1.14, -0.79]$	-0.07 $[-0.27, +0.13]$	-0.09 $[-0.60, +0.44]$	$+0.31$ $[-0.27, +0.87]$
Waiting, baseline asks	-1.00 $[-1.27, -0.73]$	-0.23 $[-0.36, -0.10]$	$+0.14$ $[-0.03, +0.32]$	$+0.01$ $[-0.59, +0.60]$	$+0.08$ $[-0.50, +0.66]$
Waiting + soft asks	-1.20 $[-1.48, -0.92]$	-1.02 $[-1.21, -0.84]$	$+0.01$ $[-0.24, +0.30]$	-0.32 $[-1.04, +0.42]$	-0.29 $[-1.06, +0.53]$

Table 2: Pooled stages A+C (90 paired episodes), difference from the supplied greedy baseline. Ask cost: -3 units for targeted asks, 0 for baseline asks, $+51.5$ for soft asks. Greedy's mean is about 0.7 msmi per episode.

Q1: does the build reproduce the prototype? On fresh seeds the prototype (networkx blossom, soft asks) versus the stdlib Round 1 build differs by -0.28 msmi $[-0.69, +0.14]$ and $+0.22$ mutual acceptances $[-0.44, +0.97]$, within noise; it does spend $+53.7$ ask units and gets $+0.89$ dates $[+0.17, +1.67]$ (one of eight outcomes, not corrected for multiplicity). At the decision level, on 337 days from 9 fresh episodes the exact stdlib matching has the same total weight as networkx on 333 days (98.8%) and the identical pair set on 285 (84.6%; ties). The largest value gap is 5.8%, and exactly four days had a component above the 18-member exact limit, as many as the unequal days.

Q2: do calibrated propensities help? Hand-set predictions of mutual acceptance average 0.10 against 0.137 observed; fitted priors give 0.118. On fresh seeds the Brier score is 0.119 (hand-set) and 0.118 (fitted), identical to predicting the base rate (0.118); AUC is 0.554 and 0.549 (0.52 on stage A). Fitted-prior versus constant-prior outcomes on fresh seeds: msmi $+0.08$ $[-0.17, +0.31]$, mutual acceptances $+0.56$ $[-0.14, +1.28]$. Hand-set learned versus constant prior, pooled: msmi $+0.13$ $[+0.02, +0.26]$, mutual acceptances $+0.16$ $[-0.28, +0.59]$, but fresh seeds alone gave $+0.17$ $[-0.06, +0.42]$ msmi and the best-looking variant was chosen among many, so I do not call it confirmed. An oracle using the simulator's true acceptance probabilities gets $+1.03$ mutual acceptances $[+0.25, +1.81]$ over greedy on fresh seeds but no gain on waiting, so ordering signal exists but is small relative to what 1–3 observations per member can learn.

Q3: changing conditions. With only 4 episodes per condition, intervals are wide. Pooled over the 20 stress episodes the waiting rule still lowers members left waiting (-0.30 $[-0.55, -0.05]$ vs greedy), and targeted constraint asks remain essential: removing all asks adds $+37.9$ $[+30.7, +44.4]$ left-waiting members and loses 5.0 mutual acceptances versus greedy. Soft asks cost about $+45$ units for $+0.05$ msmi $[0.00, +0.15]$ over the default and never helped. Learning on top of the default is $+0.05$ msmi $[-0.10, +0.20]$ overall and $+0.25$ $[0.00, +0.75]$ in each of the two conditions where members really differ (bimodal response, strong acceptance bias), the only place a gain is plausible; when response behaviour drops mid-episode it is -0.25 $[-0.75, 0.00]$. Baseline asks give $+0.65$ mutual acceptances versus the default for $+2.6$ units, again within noise. The 70%-missing world is degenerate (greedy coverage 0.04).

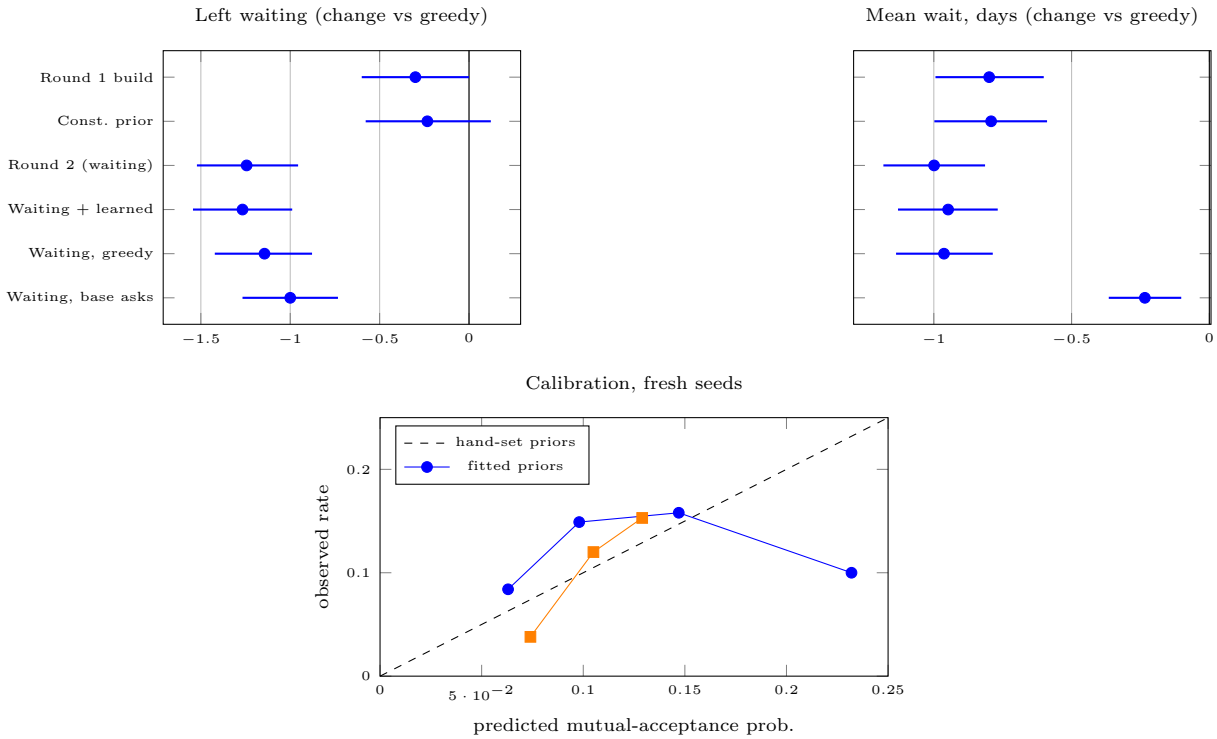


Figure 1: Top: paired differences from greedy (pooled A+C, 95% intervals); the vertical line is zero. Bottom: reliability of the mutual-acceptance prediction on fresh seeds.

6 What the comparisons taught

(i) The waiting priority does the work: exact matching adds nothing over greedy allocation with the same priority on who is left waiting (-1.24 vs -1.14), so a simpler allocator would do; I keep exact matching because it costs 0.13 s per call. (ii) Against the Round 1 build, the waiting rule is a trade: pooled, Round 1 has $+0.72$ $[+0.23, +1.20]$ mutual acceptances and $+0.67$ $[+0.20, +1.16]$ dates over Round 2 but leaves 0.94 $[0.67, 1.23]$ more members waiting; Round 2 is only "free" relative to greedy. (iii) The waiting+learned variant recovers most of both but did not beat the default on fresh seeds alone; it is the candidate for a properly powered Round 3. (iv) Targeted asks are mainly a waiting-time tool (mean wait -1.00 vs -0.23 with baseline asks), not a cost saver (-3 units). (v) msmi stays too rare (under one event per episode) to separate any of these; mutual acceptance, dates and waiting are the usable outcomes.

7 Limitations

Stage-A selection used about eleven policies, so its single positive msmi results are subject to selection; stage C and the pooled intervals are uncorrected for multiplicity. Stress runs have 4 episodes per condition. $\lambda = 2$ is an untuned choice (1, 2, 4 gave the same left-waiting result on stage A). The official evaluator was run for seed 101 only (all six scenarios valid; primary score 0.50, coverage 0.4175, mutual acceptances 8.5 per 100, ask cost 199, 8.3 s inference in total outside the container); the container build and run were checked on my machine (seed 101, all six scenarios valid, primary score 0.50, about 98 s of inference per episode, i.e. roughly 0.8 s per call including container startup, against the 10 s limit).

8 Conclusion

What the outcome is. The Round 2 policy keeps the score at the level of the supplied greedy baseline but serves people more fairly: pooled over 90 episodes it leaves about 1.2 fewer members waiting per episode despite having a feasible partner $[-1.52, -0.96]$ and cuts the average wait to a first introduction by about one day, with no detectable change in MSMI, mutual acceptances or dates against greedy. Every episode on the official evaluator and in the Docker container was valid.

What it does not show. It does not prove a higher MSMI: with under one event per episode, no MSMI difference in this study separates from zero. It also does not keep the Round 1 build's mutual-acceptance advantage; that gain and the waiting gain trade against each other.

What I would do next. (1) Run 30 or more fresh seeds and judge the key comparisons on mutual acceptances, dates and waiting, which are less noisy than MSMI. (2) Decide in advance whether waiting plus learned propensities replaces the default, because it recovered most of both gains but was chosen after seeing the data. (3) Tune the waiting weight on held-out seeds.

Why it is a sound entry. It follows the data instead of the Round 1 expectations: it states which hypotheses failed, shows that a simpler method does about as well, keeps hard eligibility and observed preferences untouched, and is fully reproducible from the included code and results.

9 Reproducibility and provenance

Python 3.12, standard library only for the policy; analysis used numpy, scipy, matplotlib and networkx (prototype only). No training; priors fixed constants or fitted on seeds 3001–3009; deterministic. See `README.md` and `scripts/run_round2.ps1`. The simulator source was read to design experiments and the oracle uses simulator truth only inside the evaluation harness. Coding and analysis assistance: Claude (Anthropic)